

CSE 214

Online on Creational Pattern

Time: 20 minutes

You are developing a configuration manager for a high-performance game engine. The game has a configuration file that contains settings like Resolution, Audio Volume, and Difficulty Level.

Since loading these settings from the disk is expensive, you want to load them once and keep them in memory. Every part of the game (Graphics engine, Audio engine, AI) needs to access these same settings. It is strictly forbidden to have multiple copies of the configuration object in memory, as this could lead to inconsistent game states.

Task: Implement the `GameConfig` class. Ensure that no matter how many times a client tries to instantiate this class, they always get the same, single instance.